Alexander Madurowicz, PhD

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POSTDOCTORAL EMPLOYMENT

Postdoctoral Researcher at Space Telescope Science Institute Fall 2024 - Present
- Application of JWST NIRSpec IFU spectroscopy to GO 1282, 2044, 6362, and 8063
  - Empirical modeling of spectral noise covariance to debias likelihood analyses
  - Supplementing atmospheric models with custom clouds and compositional perturbations
  - Development of tutorials and code optimization for BREADS software package
- Organizational lead for ESPF Science Hour weekly seminar/journal club Fall 2025 - Present

Postdoctoral Researcher at University of California, Santa Cruz. Fall 2023 - Fall 2024
Development of high contrast spectroscopy with JWST NIRSpec for GO 3522/GTO 1414

EDUCATION

Doctor of Philosophy in Physics, Stanford University, GPA: 3.9 Fall 2017 — Spring 2023
Thesis: Universal Optical Instrumentation for Exoplanet Atmospheres
Full PDF Document Video of Defense Presentation

Bachelor of Arts in Physics, University of California Berkeley, GPA 3.9 Fall 2013 — Spring 2017
Olsen's Scholarship Recipient 2015 — 2016
Deans Honors Spring 2014

FIRST AUTHOR PUBLICATIONS

The Dyn-Atmo Survey: JWST/NIRSpec spectroscopy of dynamical benchmark GJ 758 B
arxiv:2609.11828

Direct Spectroscopy of 51 Eridani b with JWST NIRSpec
The Astronomical Journal, Volume 170, Number 6 (2025)

The Infrared Colors of 51 Eridani b: Micrometeoroid Dust or Chemical Disequilibrium?
The Astronomical Journal, Volume 165, Number 6 (2023).

Integral Field Spectroscopy with the Solar Gravitational Lens
The Astrophysical Journal Volume 930 Number 1 (2022).

Resolving Exo-Continents with Einstein Ring Deconvolution
arxiv:2003.13918

GPI 2.0: Optimizing Reconstructor Performance in Simulations and Contrast Estimates
Proceedings of SPIE Volume 11448, Adaptive Optics Systems VII; 114482H (2020).

Asymmetries in adaptive optics point spread functions
J. of Astronomical Telescopes, Instruments, and Systems, 5(4), 049003 (2019).

Characterization of lemniscate atmospheric aberrations in Gemini Planet Imager Data
Proceedings of SPIE Volume 10703, Adaptive Optics Systems VI; 107036E (2018).

NOTABLE COUATHORSHIP

Impacts of Correlated Noise on Retrievals of Exo-Earth Atmospheres
The Astronomical Journal, Volume 172, Number 4 (2026)

Discovery of an Exterior Third Planet Orbiting β Pictoris
The Astrophysical Journal Letters, Volume 1006, Number 1 (2026)

JWST-TST High Contrast: Achieving Direct Spectroscopy of Faint Substellar Companions Next to Bright Stars with the NIRSpec Integral Field Unit
The Astronomical Journal, Volume 168, Number 2 (2024)

PRESENTATIONS and CONFERENCE POSTERS

The 23rd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun June 2026
Poster: "Panchromatic JWST Spectrophotometry of benchmark brown dwarf GJ 758 B"

The Brown Dwarf to Exoplanet Connection (BDEXOCON) Contributed Talk April 2026
"Direct spectroscopy of cool exoplanets and brown dwarfs with JWST / NIRSpec"

Spirit of Lyot 6 Contributed Presentation February 2026
"Direct Spectroscopy of cool exoplanets and brown dwarfs with JWST/NIRSpec"

Space Telescope Science Institute ExoJamboree Contributed Talk November 2025
"Direct Spectroscopy of cool exoplanets and brown dwarfs with JWST/NIRSpec"
Ludwig Maximillian University of Munich Galaxies and Lensing Seminar June 2024
Invited Talk "Resolving Exoplanets with the Solar Gravitational Lens"
Exoplanets 5
Poster for "Signature of Chemical Disequilibrium in the infrared colors of 51 Eridani b"
UCSC Planetary Lunch Seminar January 2024
Invited Talk: "Evidence for disequilibrium chemistry in the spectra of 51 Eridani b"
UC Berkeley Center for Integrative Planetary Science Seminar November 2023
Invited talk: "Resolving exoplanets with solar gravitational lens"
Spirit of Lyot 5 June 2022
Poster for "Integral Field Spectroscopy for the Solar Gravitational Lens"
SPIE Astronomical Telescopes and Instrumentation Adaptive Optics Systems VII 2020
Presentation on "GPI 2.0: optimizing reconstructor performance in simulations and preliminary contrast estimates"
Virtual Presentation Video
Adaptive Optics for Extremely Large Telescopes 6 Poster 2019
"Asymmetries in adaptive optics point spread functions"
SPIE Astronomical Telescopes and Instrumentation Adaptive Optics Systems VI Poster 2018
"Characterization of lemniscate atmospheric aberrations in Gemini Planet Imager Data"
Annual Meeting of the American Physical Society Presentation April 2017
"Development of an Inductively Coupled Thermometer for a Cryogenic Half-Wave Plate"
Award: "Best Undergraduate Presentation"

PRIOR RESEARCH EXPERIENCES

Graduate Research Assistantship, Stanford University Fall 2017 — Spring 2023
- Designed and executed multiple independent and collaborative research projects resulting in multiple first author publications
- Utilized distributed supercomputer resources to analyze large datasets and run expensive computational models
- Traveled to remote locations to perform astronomical observations on site

Research Assistant, Lawerence Berkeley National Laboratory Summer 2015 — Spring 2017
- Designed, built, and tested an inductive thermometer for a cryogenic half-wave plate
- Refurbished microKelvin cryogenic dewar
- Designed and built close circuit water cooling system for helium compressor

TEACHING and MENTORING

Teaching Assistantship
- Stanford University Physics 23 "Electricity, Magnetism, and Optics" Winter 2022
Standard introductory course assistantship involving hosting discussion session with practice problems, office hours for special help, and grading homework and exams
- Stanford University Physics 16 "Origin and development of the cosmos" Winter 2021
Similar assistantship with greater flexibility in designing custom course material
- Stanford University Physics 15 "Stars and Planets in a Habitable Universe" Fall 2018
Similar to Physics 15 with a different astrophysical focus

Undergraduate Mentorship For Avi Kaplan-Lipkin Summer 2021 — Summer 2022
Publication: Multiwavelength Mitigation of Stellar Activity in Astrometric Planet Detection
The Astronomical Journal, Volume 163, Number 5
Involved regular meetings, introductory test problems, literature review, sharing of programming practices, debugging, sharing insights, paper draft commentary, general knowledge transfer, and other mentorship activities.

PUBLIC OUTREACH and COMMUNICATION

Frontier Space Spotify Podcast Guest Appearance June 2022
Solar Gravitational Lens for Exoplanet Surface Imaging and Interstellar Communication
Weekly Space Hangout Youtube Podcast Guest Appearance May 2022

Bringing Exoplanets into Focus, Bit by Bit
Wonderfest "Ask A Science Envoy" Public Zoom Presentation April 2022
Imaging Extrasolar Planets
KPOO FM89.5 Radio Interview October 2021
Scintillating Science with DJ Marilynn Fowler

IN POPULAR MEDIA

Stanford News: Stanford scientists describe a gravity telescope that could image exoplanets
KIPAC Highlights: The telescope at the edge of the solar system
Scientific American: Our Sun Could Someday Reveal the Surfaces of Alien Earths
Inverse Magazine: This futuristic telescope would use Einsteinian physics to find Earth 2.0

MISCELLANEOUS ACADEMIC PROJECTS

- Exoplanetary Atmosphere Journal Club Leader Fall 2020
  Organizational leadership experience. Coordinating and preparing presentations discussing the Zhang (2020) Review, advertising the meeting cross-institutionally to interested researchers
- Comparative Topography of Terrestrial Solar System Bodies Fall 2019
  Project for GEOLSCI 222 “Planetary Systems”: Empirical analysis of the statistical distributions of elevations for all solar system bodies for which this data is available
- Limitations of the Universal Approximation Theorem Spring 2019
  Essay for APPhys 293 “Theoretical Neuroscience”: A simple proof the powerful flexibility of neural networks as approximate functions, and an elaboration around why they can still fail despite this
- Insulator-Conductor Transition in Liquid Metal Hydrogen Winter 2019
  Essay for Phys 212 “Statistical Mechanics”: Theoretical and computational estimates of the phase transition of hydrogen in pressures and temperatures relevant to the interiors of giant planets
- Interpolating Brown Dwarf Atmosphere Models with Machine Learning Fall 2018
  Project for CS 229 “Machine Learning”: Multi-linear matrix interpolation on a non-uniform grid for various atmospheric model parameters
- PSF Classification with Deep Neural Networks Winter 2018
  Project for CS 230 “Deep Learning”: Custom architecture binary classifier with hand-labeled data from astrophysical research
- The CNB is probably not the Dark Matter Fall 2017
  Essay for Phys 361 “Cosmology”: Estimates regarding the cosmic neutrino background
- Cloud Atlas Earth Visualization Summer 2017
  Personal project: geospatial data assimilation, map projections, and a raspberry pi

OTHER ASSORTED SKILLS

Python, UNIX, LaTeX, Jupyter, Tensorflow, MATLAB, Mathematica, LabVIEW, AutoCAD, Adobe Creative Suite, HTML, CSS, PHP